Cryptography
Lab Assignment

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6) Write a program find addition, subtraction, Multiplicati
// Operations +, -, *, /, %. division, modulus.

```
#include <iostream>
using namespace std;
int main(){
    //Assuming variables as integer
    int a, b, ch;
    cin>>a;
    cin>>b;
    cout<< "Enter choice for add(1)\n subtraction(2)\n Multiplication
        (3)\n Division(4)\n Modulus (5)\n";
    cin>>ch;
    try {
      if( ch<1 || ch>5 ){
          Throw " Invalid choice;
      }
    if((ch==5) || ch==4) && b==0) {
          Throw Division/ Modulus Denominator should not be
           zero",
    }
switch(ch) {
    case 1:
        cout<< "Addition :: "<<a+b<<endl;
        break;
    case 2:
        cout<< "Subtraction:<< a-b << endl;
        break;
    case 3:
        cout<< "Multiplication << a*b <<endl;
        break;
    Case 4:
        cout<< "Division<< a/b<< endl;
        break;
    case 5;
      cout << "Modulus << a%b <<endl;
      break;
}
```

```
catch (const char* msg) {
    cout << "Exception caught :: " << msg << endl;
}
return 0;
}
```

Output: 5
4
1

Addition :: 9

Test Cases

	a	b	ch	Output
1.	0	0	1	Addition: 0
2.	1	1	0	Exception: Invalid choice
3.	2	3	2	Subtraction: -1
4.	-3	4	2	Subtraction: -7
5.	-3	-3	3	Multiplication: 9
6.	4	0	4	Exception: Division / Modulus Denominator should not be zero
7.	-5	-5	-5	→ Exception: Invalid choice
8.	0	0	0	Exception: Invalid choice
9.	1	1	1	Addition: 2

hon Online Compiler

```python
try:
    a = int(input("Enter first number: "))
    b = int(input("Enter second number: "))

    print("\nChoose an operation:")
    print("1. Addition")
    print("2. Subtraction")
    print("3. Multiplication")
    print("4. Division")
    print("5. Modulus")

    ch = int(input("Enter your choice: "))

    if ch < 1 or ch > 5:
        raise Exception("Invalid Choice")

    if (ch == 4 or ch == 5) and b == 0:
        raise ZeroDivisionError("Division/Modulus denominator should not be zero")

    if ch == 1:
        print("Addition:", a + b)
    elif ch == 2:
        print("Subtraction:", a - b)
    elif ch == 3:
        print("Multiplication:", a * b)
    elif ch == 4:
        print("Division:", a / b)
    elif ch == 5:
        print("Modulus:", a % b)

except ZeroDivisionError as e:
    print("Exception Caught:", e)

except Exception as e:
    print("Exception Caught:", e)
```

```
Enter first number: 2
Enter second number: 3

Choose an operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus
Enter your choice: 2
Subtraction: -1

=== Code Execution Successful ===
```

```
Enter first number: -3
Enter second number: 4

Choose an operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus
Enter your choice: 2
Subtraction: -7
```

```java
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        try {
            System.out.print("Enter first number: ");
            int a = sc.nextInt();

            System.out.print("Enter second number: ");
            int b = sc.nextInt();

            System.out.println("\nChoose an operation:");
            System.out.println("1. Addition");
            System.out.println("2. Subtraction");
            System.out.println("3. Multiplication");
            System.out.println("4. Division");
            System.out.println("5. Modulus");

            System.out.print("Enter your choice: ");
            int ch = sc.nextInt();

            if (ch < 1 || ch > 5) {
                throw new Exception("Invalid Choice");
            }

            if ((ch == 4 || ch == 5) && b == 0) {
                throw new ArithmeticException("Division/Modulus denominator should not be zero");
            }

            switch (ch) {
                case 1:
                    System.out.println("Addition: " + (a + b));
                    break;
                case 2:
                    System.out.println("Subtraction: " + (a - b));
                    break;
                case 3:
                    System.out.println("Multiplication: " + (a * b));
                    break;
```

```
            if (ch < 1 || ch > 5) {
                throw new Exception("Invalid Choice");
            }

            if ((ch == 4 || ch == 5) && b == 0) {
                throw new ArithmeticException("Division/Modulus denominator should not be zero");
            }

            switch (ch) {
                case 1:
                    System.out.println("Addition: " + (a + b));
                    break;
                case 2:
                    System.out.println("Subtraction: " + (a - b));
                    break;
                case 3:
                    System.out.println("Multiplication: " + (a * b));
                    break;
                case 4:
                    System.out.println("Division: " + (a / b));
                    break;
                case 5:
                    System.out.println("Modulus: " + (a % b));
                    break;
            }

        } catch (ArithmeticException e) {
            System.out.println("Exception Caught: " + e.getMessage());
        } catch (Exception e) {
            System.out.println("Exception Caught: " + e.getMessage());
        }

        sc.close();
    }
```

```
Enter first number: 0
Enter second number: 0

Choose an operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus
Enter your choice: 1
Addition: 0

=== Code Execution Successful ===
```

```
Enter first number: 1
Enter second number: 1

Choose an operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus
Enter your choice: 0
Exception Caught: Invalid Choice

=== Code Execution Successful ===
```